Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

SST TAC Production Centre

Climate Change Service

climate.copernicus.eu

Copernicus Climate Change ServiceL4 SST/IST analyses

European Space Agency Sea Surface Temperature Climate Change Initiative L4 SST analyses

SST-GLO-SST-L4-REP-OBSERVATIONS-010-024

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Contributors: I. Karagali, S. Good.

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table.

The third column specifies which sections or sub-sections have been updated. The fourth column should mention the reason for updating the document. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

Issue	Date	§	Description of Change	Author	Validated By
1.0	January 2017	All	Creation of the document	Simon Good	
1.1	March 2017		Responses to review; addition of logos and licensing information.	Simon Good	
2.0	12 April 2019	All	Update ESA SST CCI analyses to version CDR2.0	Simon Good	
2.1	10 September 2019		Added C3S ICDR2.0	Simon Good	Mercator Ocean
2.2	10 September 2020		Extension to 2019 and upstream data change	Simon Good	Bruno Buongiorno Nardelli
2.3	15 January 2021	II	Temporal extensions	Simon Good	Bruno Buongiorno Nardelli
3.0	3 October 2024	All	Replacement product	Ioanna Karagali	A. Pisano
4.0	18 July 2025		Reintroduced dataset ESACCI-GLO-SST-L4-REP-OBS-SST	Simon Good	A. Pisano
5.0	12 November 2025	All	New document template and update ESA SST CCI analyses to version CDR3.0	Simon Good	A. Pisano

ACRONYM TABLE

Acronym	Description
(A)ATSR	(Advanced) Along Track Scanning Radiometer
AASTI	Arctic and Antarctic Surface Temperatures from thermal Infrared satellite radiometers
AMSR	Advanced Microwave Scanning Radiometer
Argo	Array for Real-Time Geostrophic Oceanography. International program that measures water properties across the world's ocean using a fleet of robotic instruments that drift with the ocean currents and move up and down between the surface and a mid-water level
AVHRR	Advanced Very-High Resolution Radiometer
C3S	Copernicus Climate Change Service
CCI	Climate Change Initiative
CDR	Climate Data Record
DMI	Danish Meteorological Institute
EAN	Estimated Accuracy Number
ESA	European Space Agency
ICDR	Interim Climate Data Record
IST	Ice Surface Temperature
MAD	Median Absolute Deviation
OSTIA	Operational Sea Surface Temperature and Ice Analysis
OI	Optimal Interpolation
RSD	Robust Standard Deviation
SLSTR	Sea and Land Surface Temperature Radiometer
SST	Sea Surface Temperature
T2m	Temperature at 2m height
TAC	Copernicus Marine Service Thematic Assembly Centre

KEY DEFINITIONS

Foundation SST	The SST from which the diurnal cycle grows each day[1]
MAD	Median absolute deviation from the median
Robust standard deviation (RSD)	1.4826 × MAD

[1] https://www.ghrsst.org/ghrsst-data-services/for-sst-data-users/products/#section5

RELEVANT DOCUMENT LIST

RD.1	Karagali, I. et al. (2024). Global L4 Sea and Sea Ice Surface Temperature Climate Data Record, Version 1.0: Product User Guide and Specifications. Copernicus Climate Change Service. Document ref. C3S2-312b_WP1-FDDP-SST/IST_2024_PUGS_v1.X[2]
RD.2	Karagali, I. et al. (2024). Global L4 Sea and Sea Ice Surface Temperature Climate Data Record, Version 1.0: Algorithm Theoretical Basis Document. Copernicus Climate Change Service. Document ref. C3S2-312b_WP1-FDDP-SST/IST_2024_ATBD_v1.X[3]
RD.3	Karagali, I. et al. (2024). Global L4 Sea and Sea Ice Surface Temperature Climate Data Record, Version 1.0: Product Quality Assurance Document. Copernicus Climate Change Service. Document ref. C3S2-312b_WP1-PDDP-SST/IST_2024_PQAD_v1.2
RD.4	Karagali, I. et al. (2024). Global L4 Sea and Sea Ice Surface Temperature Climate Data Record, Version 1.0: Product Quality Assessment Report. Copernicus Climate Change Service. Document ref. C3S2-312b_WP1-FDDP-SST/IST_2024_PQAR_v1.X[4]
RD.5	ESA SST CCI Product User Guide (https://climate.esa.int/media/documents/SST_CCI_D4.2_PUG_v1.1.pdf)
RD.6	ESA SST CCI Product Validation and Intercomparison Report (https://climate.esa.int/media/documents/SST_CCI_D4.1_PVIR_v2.1-signed.pdf)
RD.7	C3S Product Quality Assessment Report (https://dast.copernicus-climate.eu/documents/satellite-sea-surface-temperature/v2.1/D2.SST.2-v3.0_PQAR_of_v2.1SST_products_v5.1_APPROVED_Ver1.pdf)

2 Users are advised to consult the latest version of the PUGS document available.
3 Users are advised to consult the latest version of the ATBD document available.
4 Users are advised to consult the latest version of the PQAR document available.

TABLE OF CONTENTS

Acronym Table *3*

Key definitions *3*

Relevant document List *4*

I Executive summary *6*

I.1 Products covered by this document **6**

I.2 Summary of the results **6**

I.2.1 Summary of results for the C3S analyses 6

I.2.2 Summary of results for the ESA SST CCI analyses 6

I.3 Estimated Accuracy Numbers **6**

II Production system description *8*

II.1 C3S analyses **8**

II.1.1 Production centre name 8

II.1.2 Description 8

II.2 ESA SST CCI analyses **8**

II.2.1 Production centre name 8

II.2.2 Description 8

III Validation framework *10*

IV Validation results *11*

IV.1 Validation results for C3S analyses **11**

IV.2 Validation results for ESA SST CCI analyses **11**

V System's Noticeable events, outages or changes *12*

VI Quality changes since previous version *13*

VI.1 Quality changes since the previous version of the C3S analyses **13**

VI.2 Quality changes since the previous version of the ESA SST CCI analyses **13**

VII References *14*

I EXECUTIVE SUMMARY

I.1 Products covered by this document

The product SST_GLO_SST_L4_REP_OBSERVATIONS_010_024 consists of two datasets of sea surface temperature (SST) analyses:

1) The Copernicus Climate Change Service (C3S) SST/IST analyses, which are global analyses of the sea surface temperature at 20 cm depth and sea-ice surface temperature (skin), representing approximately the daily average. The product is described fully in [RD.1] and [RD.2].
2) The European Space Agency Sea Surface Temperature Climate Change Initiative (ESA SST CCI) SST analyses, which are global analyses of the sea surface temperature at 20 cm depth, representing approximately the daily average. The product is described fully in Embury et al. (2024) and [RD.5].

I.2 Summary of the results

For C3S analyses, the applied validation methodology is described in detail in [RD.3] while the full validation results will be presented in [RD.4]. For ESA SST CCI analyses, the methodology and quality are described in [RD.6]. In summary, robust statistics of differences between the analyses and reference datasets are used to assess both the sea surface and sea-ice surface temperatures and their uncertainties.

I.2.1 Summary of results for the C3S analyses

The median difference between the SST component of the analyses and drifting buoy reference data ranged from -0.04 to 0.02 K, depending on the period and reference data subset that was used. Robust standard deviation ranged from 0.23 to 0.30 K.

The median difference between T2m (near-surface air temperature, typically around 2 m height) and IST ranged from -2 to -3 K depending on the period, data set and hemisphere. Robust standard deviation ranged from 3 to 3.5 K.

In general, the analysis uncertainties were found to be well represented.

Validation of the IST uncertainties is constrained by the lack of reference in situ observations and was therefore not performed.

I.2.2 Summary of results for the ESA SST CCI analyses

The median difference between the analyses and in situ reference data ranged from -0.07 to -0.03 K, depending on the period examined. Robust standard deviation decreased from 0.50 K in the 1980s to 0.22 K in the 2010s. Aggregating the statistics across the full record revealed a median difference and standard deviation of -0.04 K and 0.25 K respectively.

In general, the analysis uncertainties were found to be well represented.

I.3 Estimated Accuracy Numbers

Table 1 lists the estimated accuracy numbers (EANs) for the product. Please also see the C3S Product Quality Assurance Documents [RD.3], the Product Quality Assessment Reports [RD.4], and the ESA SST CCI Product Validation and Intercomparison Report [RD.6] for full information on the applied methodology and accuracy of the products.

Table 1. Estimated accuracy numbers for SST from C3S and ESA CCI, and for IST from C3S, obtained by comparing the analyses to reference in situ data.

Data	**Median difference to reference data (K)**	**Robust standard deviation of differences to reference data (K)**
C3S SST	-0.04 to 0.02	0.23 to 0.30
C3S IST	-2 to -3	3 to 3.5
ESA CCI SST	-0.04	0.25

II PRODUCTION SYSTEM DESCRIPTION

II.1 C3S analyses

II.1.1 Production centre name

Danish Meteorological Institute (DMI) as part of another contract to the C3S known as C3S 312_b ECV (2022/C3S2_312b_Moi/SC1).

II.1.2 Description

The C3S CDR/ICDR product is a global, combined satellite SST and IST analysis created using the DMIOI (DMI Optimal Interpolation) system. The CDR is generated using input data from the Along Track Scanning Radiometer (ATSR), Advanced Very-High Resolution Radiometer (AVHRR), Sea Land Surface Temperature Radiometer (SLSTR) and the Advance Microwave Sensing Radiometer 2/E from the ESA CCI and C3S projects for the Sea Surface Temperature (SST) component. For the Ice Surface Temperature (IST) component, data from the Arctic and Antarctic ice Surface Temperatures (IST) from thermal Infrared satellite climate data record version 2.1 (AASTI v2.1) and C3S ICDR are used. It can be accessed through the Copernicus Marine Service in addition to the dissemination methods provided by the originators. Its key features include its stability and its independence from in situ data for the recent period.

The product is available from 1st January 1982 to 31st December 2024 with a foreseen 3-4 months delivery behind real time on a global regular grid at 0.05° resolution. This reprocessed analysis product provides an estimate of the SST at 20 cm depth and IST skin surface temperature.

This product is distinct from the near real time and reprocessed SST analyses produced as part of the Copernicus Marine Service, which provide only estimates of the foundation SST (no IST included) and use more input data sources to provide the best analysis globally on any single day.

A full user guide is provided in [RD.1] while additional information on the product is contained in [RD.2].

Please note that the products are made available under a Creative Commons Attribution license[5]. Please cite [RD.1] and the dataset DOI if using the data. See the product user manual for full details.

II.2 ESA SST CCI analyses

II.2.1 Production centre name

Met Office (as part of the ESA SST CCI project)

II.2.2 Description

The ESA SST CCI reprocessed analysis product is a satellite SST analysis created using the OSTIA (Operational SST and Sea Ice Analysis) system from re-processed (A)ATSR and AVHRR data as part of the ESA SST CCI project. It can be accessed through the Copernicus Marine Service in addition to the dissemination methods provided by the originators. Its key features include its stability and its independence from in situ data for the recent period.

[5] https://creativecommons.org/licenses/by/4.0/

The product is available from 1st January 1980 to 31st December 2021 on a global regular grid at 0.05° resolution. This reprocessed analysis product provides an estimate of the SST at 20 cm. The inputs to the system are SSTs at 10:30 am and pm local time which means that the analyses roughly correspond to the daily average SST.

As with the C3S analyses, this product is distinct from the near real time and reprocessed SST analyses produced as part of Copernicus Marine Service, which provide estimates of the foundation SST and use more input data sources to provide the best analysis globally on any single day.

Additional information on the product is contained in Embury et al. (2024). A full user guide is provided in [RD.5].

Please note that the products are made available under a Creative Commons Attribution license[6]. Please cite [RD.5] and the dataset DOI if using the data. See the product user manual and the ESA SST CCI terms and conditions[7] for full details.

[6] https://creativecommons.org/licenses/by/4.0/

[7] https://artefacts.ceda.ac.uk/licences/specific_licences/esacci_sst_terms_and_conditions_v2.pdf

III VALIDATION FRAMEWORK

The data were comprehensively assessed as part of the C3S and ESA SST CCI projects. Full reports of the product validation and quality assessment methodology are available [RD.3; RD.6]. Results presented here are summary statistics.

Products were evaluated against independent drifting buoys, tropical moored buoys, Argo and ship-borne radiometers for the SST component. For the IST component and due to lack of in situ measurements of the sea-ice surface temperature, a series of buoys and drifting stations were utilised, typically recording 2-meter air temperatures.

Note that assessments reported here are made using robust statistics comparing the data to the reference observations i.e. the median and robust standard deviation are used instead of mean and standard deviation. The robust measures are used because they are not as affected by the presence of outliers as the alternatives.

IV VALIDATION RESULTS

IV.1 Validation results for C3S analyses

Summary statistics show that the median difference between the SST component of the analyses and drifting buoy reference data ranged from -0.04 to 0.02 K, depending on the period and reference data subset that was used. Robust standard deviation ranged from 0.23 to 0.30 K.

For the IST component of the analysis, validation is challenging due to the lack of in situ sea-ice surface temperature observations. Thus, near-surface air temperature (typically around 2 m height, T2m) were used instead. The median difference between T2m and IST ranged from -2 to -3 K depending on the period, data set and hemisphere. Robust standard deviation ranged from 3 to 3.5 K.

Uncertainty estimates of the SST component were validated by comparing the spread of the differences between the analysis data and the reference data to the spread expected from the combination of the uncertainty in the reference data and the analysis uncertainties. This was done for different levels of analysis uncertainty in order to check whether the uncertainties were well represented across the range of possible values. In general, the analysis uncertainties were found to be well represented.

Validation of the IST uncertainties is constrained by the lack of reference in situ observations and was therefore not performed.

IV.2 Validation results for ESA SST CCI analyses

Summary statistics show that the median difference between the analyses and in situ reference data ranged from -0.07 to -0.03 K, depending on the period examined, as shown in Table 2. Robust standard deviation decreased from 0.50 K in the 1980s to 0.22 K in the 2010s. Aggregating the statistics across the full record revealed a median difference and standard deviation of -0.04 K and 0.25 K respectively.

Table 2. Validation statistics for the ESA SST CCI dataset, from [RD.6].

Period	Median difference to reference data (K)	Robust standard deviation of differences to reference data (K)
1980 – 1989	-0.07	0.50
1990 – 1999	-0.03	0.35
2000 – 2010	-0.03	0.23
2010 - 2019	-0.05	0.22
All	-0.04	0.25

Uncertainty estimates were validated as described for the C3S analyses. In general, the analysis uncertainties were found to be well represented. The results do indicate that either the L4 uncertainties or the assumed in situ uncertainties are too high, but [RD.6] states that the latter is thought to be true.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section summarises the changes that have occurred in the operational systems over time (Table 3).

Table 3. Changes that have occurred in the operational system.

Date	Change/Event description	System version	Other
July 2019	Update to ESA SST CCI version CDR2.0.	CDR2.0	
December 2019	Add C3S ICDR2.0	ICDR2.0	
December 2020	Add 2019 data. Analyses for 2019 are based on data from the AVHRR sensor on MetOp-A and SLSTR on Sentinel 3A and 3B (for comparison, 2018 was based on AVHRRs on NOAA-19 and MetOp-A and SLSTR on Sentinel 3A).	ICDR2.0	
October 2024	Release of replacement product combining SST and IST and using ESA SST_cci v3.0 L3U input data	CDR1.0	1982-2021
October 2024	Release of replacement product combining SST and IST and using C3S/EOCIS/MCAS ICDR L3U input data	ICDR1.0	2022-2023
January 2025	Extension of ICDR	ICDR1.0	2024
March 2026	Update of ESA SST CCI data to version 3.0	CDR3.0	1980-2021

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section provides a short summary of the product changes between the previous and current version.

VI.1 Quality changes since the previous version of the C3S analyses

CDR: Data covers 01/1982 to 12/2021

ICDR: Data covers 01/2022 onwards

VI.2 Quality changes since the previous version of the ESA SST CCI analyses

Data now covers 01/1980 to 12/2021, compared to 09/1981 – 12/2016 in the previous version.

The dataset quality has improved since the last version due to better treatment of biases in the generation of the data used by the analysis system and inclusion of extra input data. Improvement in quality is evidenced by the change in robust standard deviation of the product relative to in situ reference data, which is 0.25K for the latest version compared to 0.27K (evaluated against drifting buoy data) in the previous version [RD.7].

VII REFERENCES

Embury, O., Merchant, C.J., Good, S.A., Rayner, N.A., Høyer, J.L., Atkinson, C., Block, T., Alerskans, E., Pearson, K.J., Worsfold, M., McCarroll, N., Donlon, C. Satellite-based time-series of sea-surface temperature since 1980 for climate applications. Scientific Data 11, 326 (2024). https://doi.org/10.1038/s41597-024-03147-w